UNIVERSITÄT BONN

Chapter 9:
Machine Learning based on Neuroimaging

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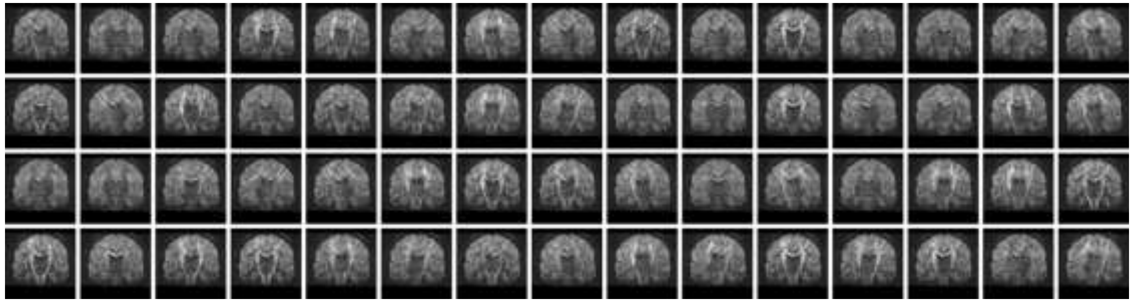
b-it Bonn-Aachen International Center for Information Technology

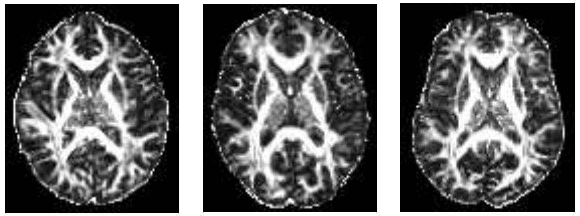
Image Acquisition and Analysis in Neuroscience

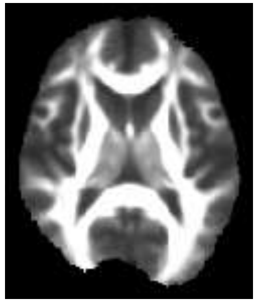
Acquisition

Reconstruction & Artifact Correction



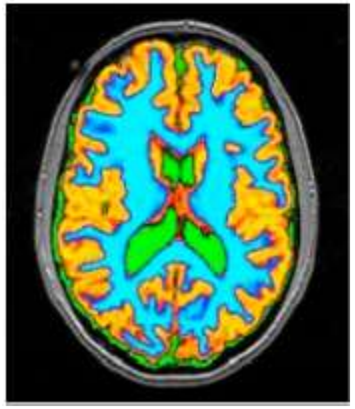
Registration



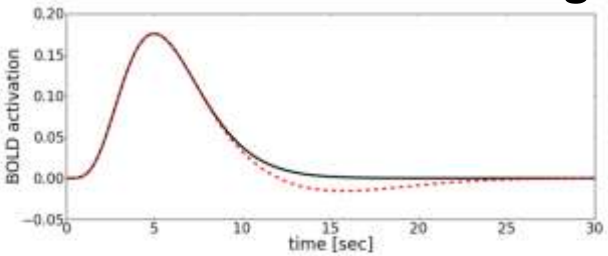


Atlas Construction

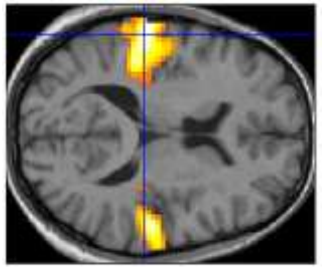
Segmentation

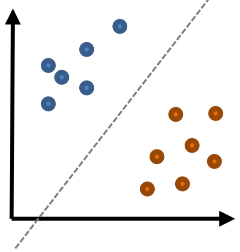


Mathematical Modeling



Hypothesis Testing





Machine Learning

Motivation: From Group to Subject

- So far, we showed uses of ML for reconstruction, registration, segmentation, tractography, but focused on **group comparisons**
- Now: Given a brain scan from one **individual subject,** can we tell...
  - whether they suffer from a particular brain disease?
  - how old they are?
  - what they saw/heard/thought of while the scan was taken?
- We will focus on **applications**
  - Some technical details (e.g., foundations of neural networks) have been introduced before
  - Others are left to introductory lectures such as MA-INF 4111 Principles of Machine Learning

Images: [Takagi/Nishimoto, 2023]

9.1 Technical Background

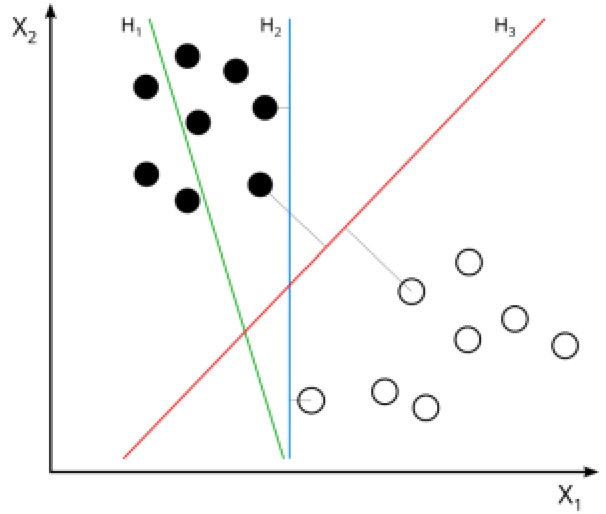
Reminder: Supervised Classification and Regression

Supervised classification and regression aim to learn a function $y = f(\mathbf{x})$ that infers a **label** *y* from a **feature vector x** whose coefficients are the individual **features**

- **Training data:** Examples $\{\mathbf{x}_i, y_i\}$ used to learn the function f
- **Classification:** y is discrete
  - *Special case:* Binary classification with $y = \pm 1$
  - *Example:* Given a brain scan, classify between MCI and Alzheimer
- **Regression:** y is continuous
  - *Example*: Given a brain scan, predict (brain) age of the person

Key assumption in classical machine learning: All observations are independent samples from a fixed probability distribution $P(\mathbf{x}, y)$

Technical Background: Support Vector Machines



- **Support vector machines** perform linear binary classification with maximal margins
  - Margin can be expressed by an often small set of “support vectors”
- Most practical uses extend this idea by
  - Learning to ignore label noise **(“soft margin”)**
  - Nonlinear classification by mapping to a higher-dimensional feature space (implicitly, **“kernel trick”**)
- Variants of this idea can also perform **regression** and **one-class classification** / anomaly detection

Image: Zack Weinberg via Wikimedia

Why Support Vector Machines?

Reasons for the popularity of **Support Vector Machines** in neuroimaging applications:

- Deal relatively well with few samples in high-dimensional spaces
- Lead to a quadratic optimization problem (unique solution, efficient if training data is not too big)
- Only retain part of the training data (explicit abstraction)
- Flexible with respect to task (e.g., classification, regression) and similarity measure (kernel)

Motivation: Feature Selection

- **Reminder:** A *feature* is one data attribute (coefficient of **x**) that will be used as input to the learning machine
- In neuroimaging, the number of features can be very high
  - E.g., value at each voxel as a feature
- Many features correlate poorly with the desired output *y*
  - Including them reduces the performance of classification or regression methods
- **Feature selection / scaling** aims to remove / reduce the impact of unimportant features
  - Also improves interpretability

Feature Selection Strategies

1. **Embedded Methods**
   – Integrate feature selection into the training process (e.g., achieve a sparse weight vector by further regularizing a linear SVM)
2. **Wrapper Methods**
   – Try to find the optimal subset of features *for a specific learning machine*
   – Typically involves repeated training on different candidate subsets and evaluation of result
3. **Filter Methods**
   – Select a subset of features *without referring to a specific learning machine*
   – Try to assess how much information features provide about labels

Summary: Technical Background

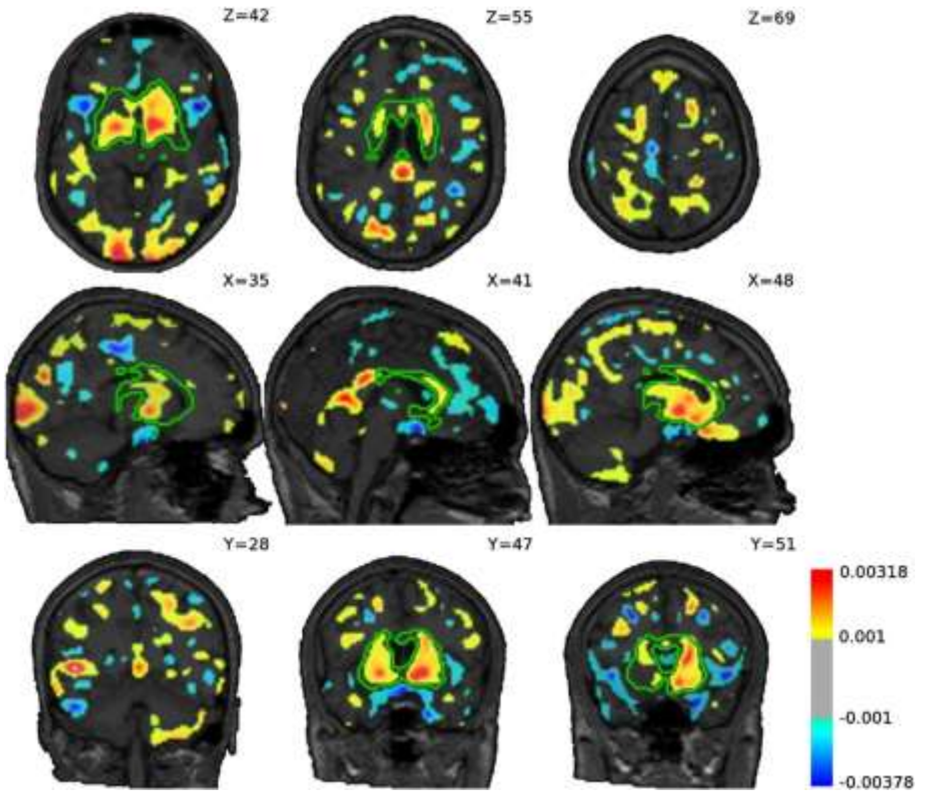
The usual **pipeline for supervised learning** from neuroimaging data involves

- Preprocessing (analogous to statistical analysis)
- Feature extraction
  - Might involve segmentation, computation of connectivity graphs, etc.
- Feature selection
  - Elimination of redundant / non-informative features
- Training of classifier / regressor
  - Classic techniques such as support vector machines still popular due to their sample efficiency

9.2 Applications in Brain Diseases

Example: Diagnosing Schizophrenia

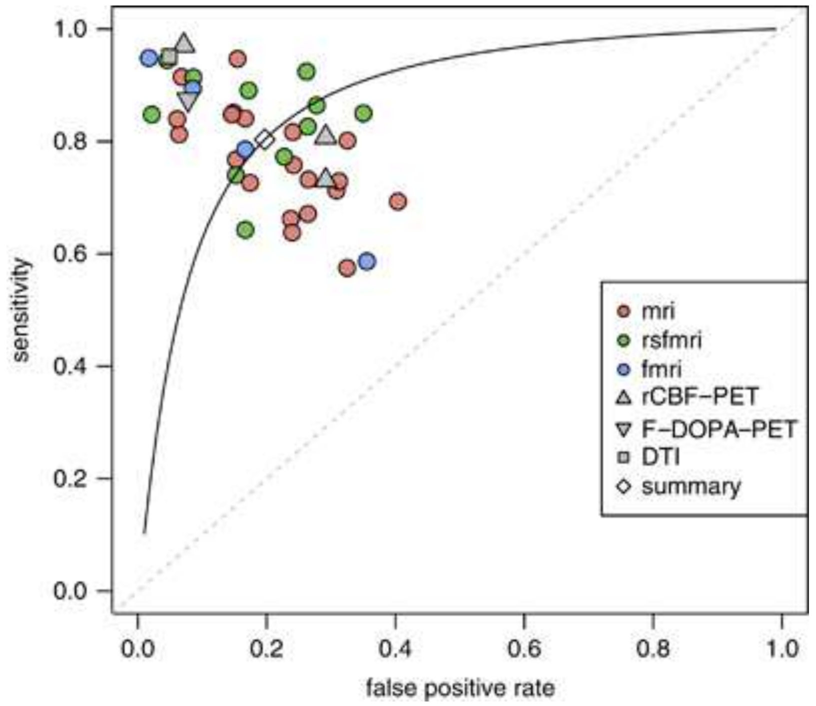
- **Nieuwenhuis et al. 2012** built a supervised classifier of patients with schizophrenia vs. healthy controls on 239 subjects, validated on 277 subjects from different scanner
  - *Features*: Gray matter density maps from VBM
    - effects of age, sex, and handedness removed via linear regression
    - downsampled to 157k features
  - *Classifier:* Linear SVM
  - *Feature selection:* Top 10% based on $\mathbf{w}$ coefficient of an initial SVM
  - Achieved 70% accuracy in test data



Visualization of **w** vector

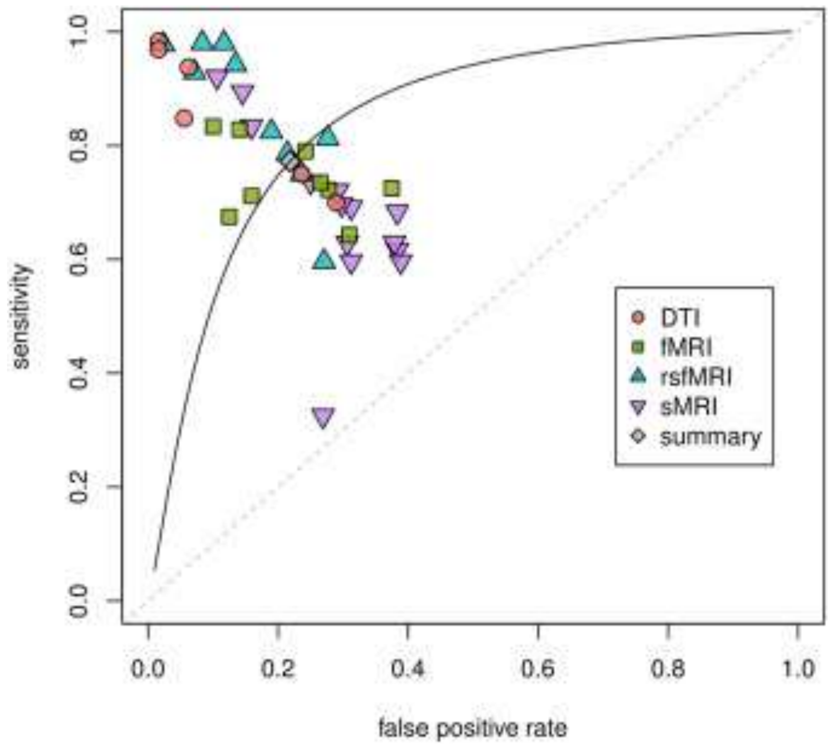
Meta Analysis: Diagnosing Schizophrenia

- **Kambeitz et al. 2015** perform a meta-analysis of 38 studies that have tried to detect schizophrenia based on structural / functional / diffusion MRI, or PET
  - Discriminant analysis and SVMs most popular
  - Achieved 80% sensitivity and specificity on average
  - rs-fMRI had higher sensitivity than structural MRI
  - Suggest that future work should combine both



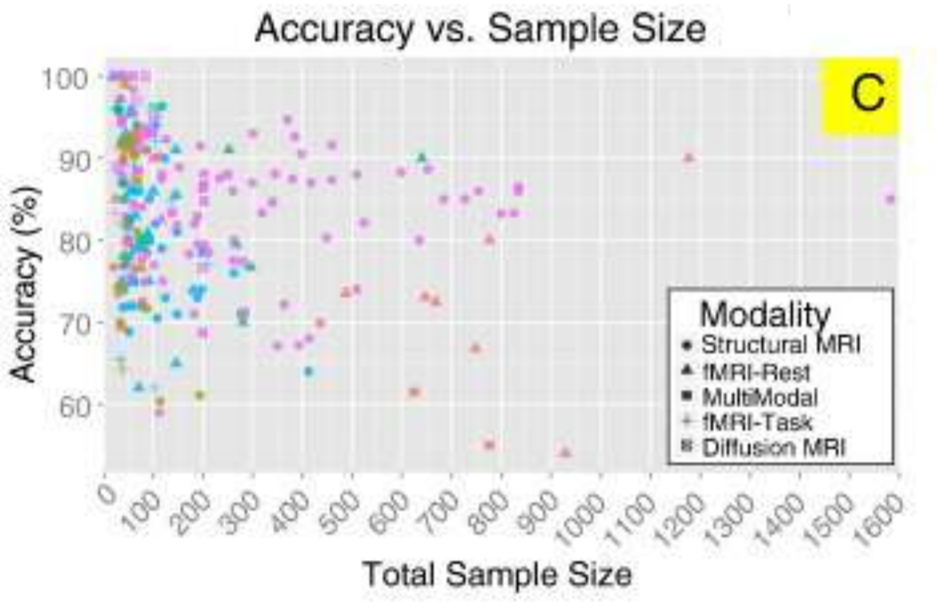
Meta Analysis: Diagnosing Depression

- **Kambeitz et al. 2016** perform a meta-analysis of 33 studies that have tried to detect major depression based on structural, functional, or diffusion MRI
  - SVM was by far the most popular classifier
  - Achieved 77% sensitivity, 78% specificity on average
  - rs-fMRI and DTI outperform structural MRI and task fMRI
  - Stress future importance of differential diagnosis



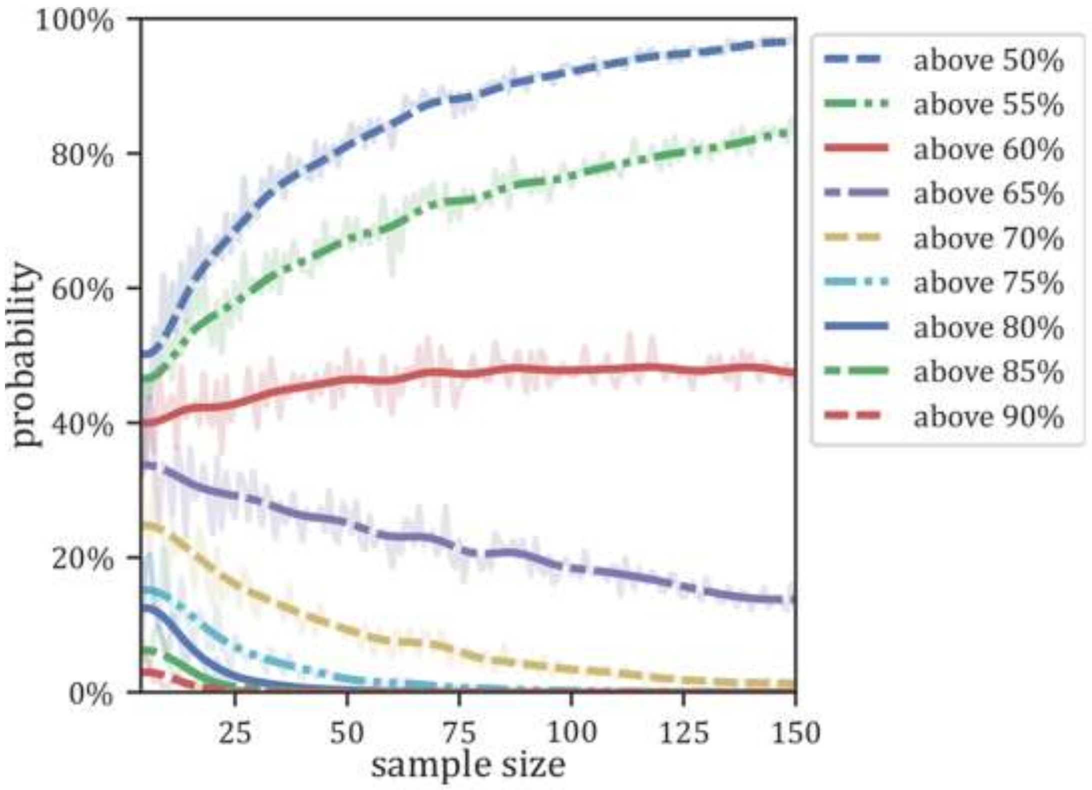
Limitations of Classifying Brain Disorders (2017)

- **Arbabshirani et al. 2017** survey more than 200 uses of machine learning for detecting schizophrenia, MCI, Alzheimer’s, depression, autism, attention-deficit hyperactivity disorder
  - Conclusion: “Immature and not ready for integration into clinical healthcare”
  - Identify sample sizes used in most studies as a limiting factor
  - Call for data sharing or methods that can work on decentralized data



Simulating the Effect of Limited Sample Size

- **Flint et al. 2021** investigate the paradox that, even though classifiers are expected to improve when trained on more data, larger studies often report lower accuracies
  - Achieved 61% accuracy in predicting major depression in a balanced sample of N=1868
  - Demonstrate that leave-one-out cross-validation on small subsamples frequently overshoots this
  - Emphasize importance of adequately sized test sets



Challenges in Classifying Mental Health Disorders (2025)

- **Dino and Hassan 2025** analyze 68 studies from 2020-2025
  - Conclude that multimodal approaches outperform unimodal ones, but "significant technical and practical challenges persist"
  - As the premier challenge (among many others), they identify dataset limitations, such as small sample sizes

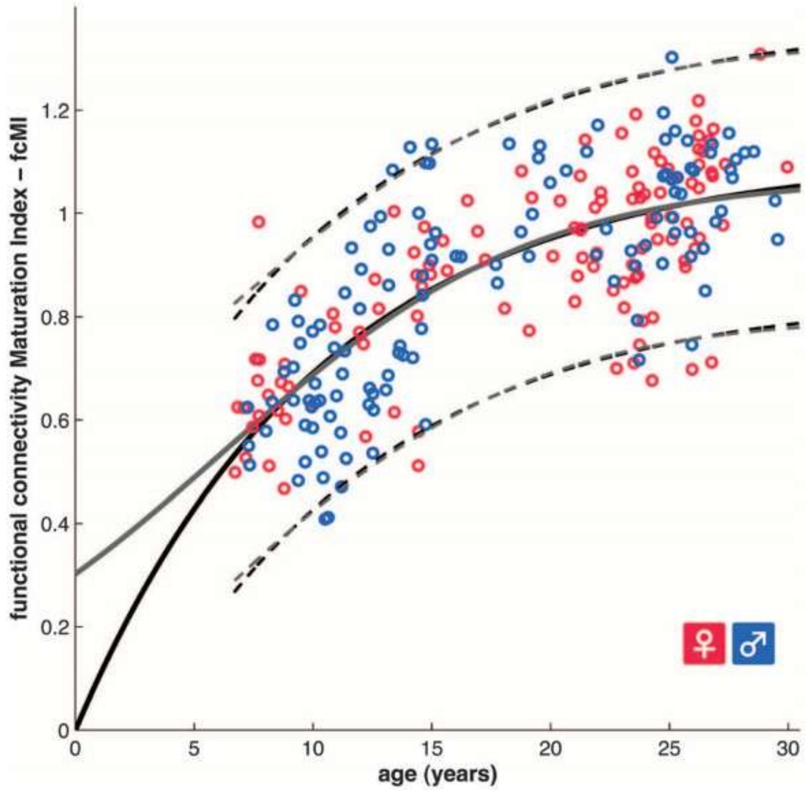
Overcoming Sample Size Limitations

- Integrating data from different centers could help overcome sample size limitations, but introduces issues with data heterogeneity
- Large-scale single-center studies such as the **Rhineland study (DZNE Bonn)** collect homogeneous and high-quality data

Image: https://www.rheinland-studie.de/

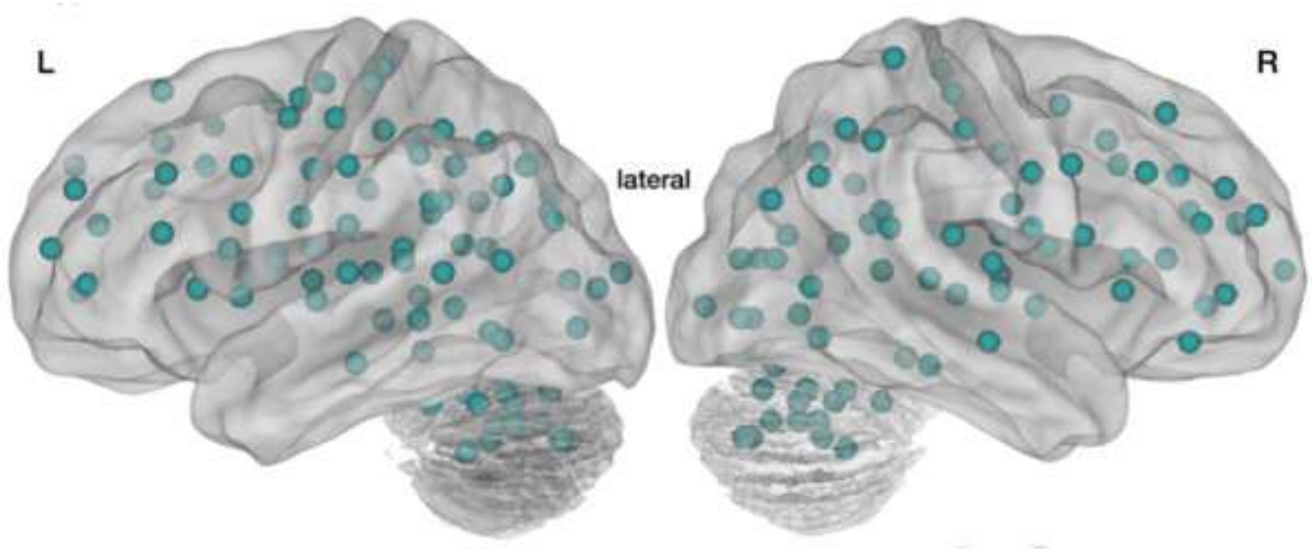
Assessing Brain Maturity

- **Dosenbach et al. 2010** predict brain maturity from rs-fMRI using SVMs
  - Five-minute scans of 238 subjects, 7-30 years
  - Classified 61 children (7-11 years) vs. 61 adults (24-30 years), matched for brain volume, with 91% accuracy
  - Derived “maturation index” from regression by normalizing to average value in range 18-30 years



Brain Maturity: Preprocessing

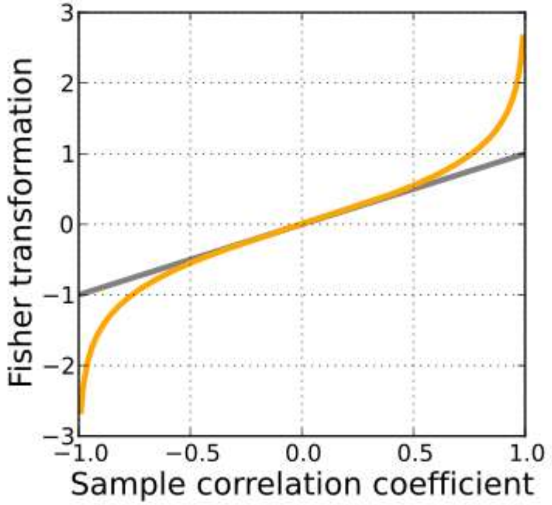
1. Standard fMRI pre-processing (Chapter 7)
2. Removal of nuisance parameters by linear regression
   - Head motion
   - Whole brain, ventricle, white matter average signal
   - First-order derivatives of those averages
3. Signal averaged over 160 predefined regions of interest (ROIs)
   - Correspond to functional areas



Brain Maturity: Feature Selection

- Compute 12,270 correlation coefficients r_i between time courses in 160 ROIs
- Take z transform to produce feature vector **x:**

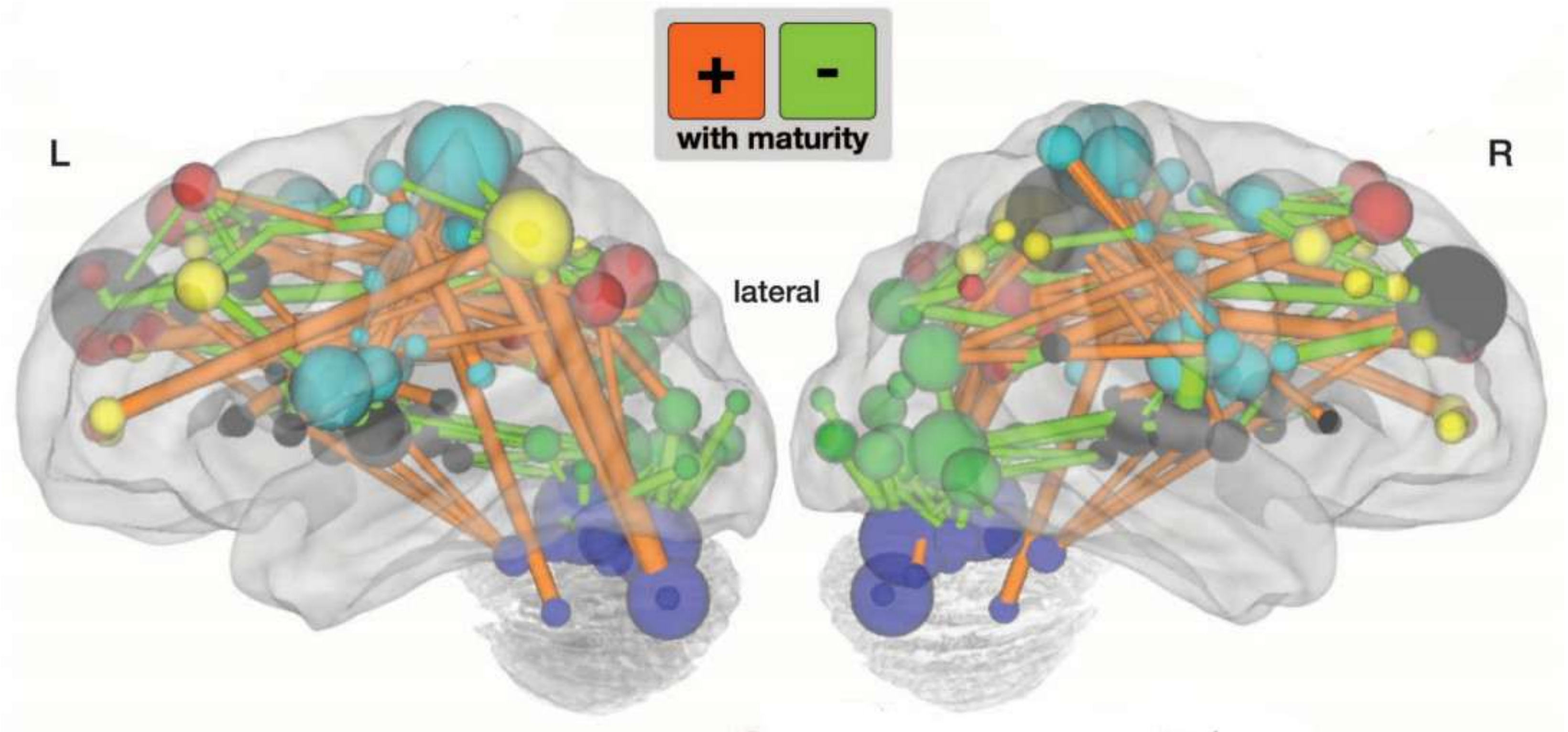
$$x_i = \frac{1}{2}\ln\frac{1+r_i}{1-r_i}$$



- Rank features by t score (for classification) or correlation with age (for regression)
  - Only retain top 200 features
  - Based on training data only

Brain Maturity: Visualization

- Visualizing **frequently selected features** (corresponding to correlations between ROIs) provides additional insight on how the brain changes with age
  - Long-range correlations tend to become stronger with age

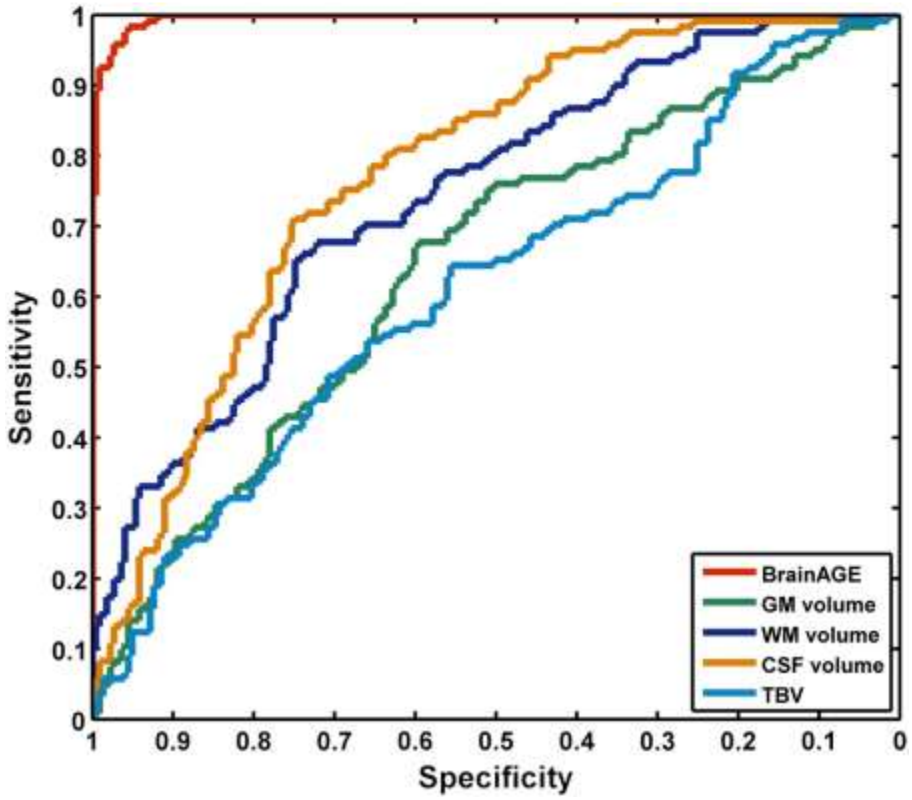


Brain Maturity: Validation

- Classification **replicated** on data from two independent fMRI studies (195/186 subjects, leave-one-out accuracy 92%/93%)
- A **permutation test** showed that classification result was significant with $p<0.0001$
- Selected features **overlap considerably** between cross-validation runs
- Visualizing selected features provides a **plausible explanation** for the basis of classification / regression

Introducing BrainAGE

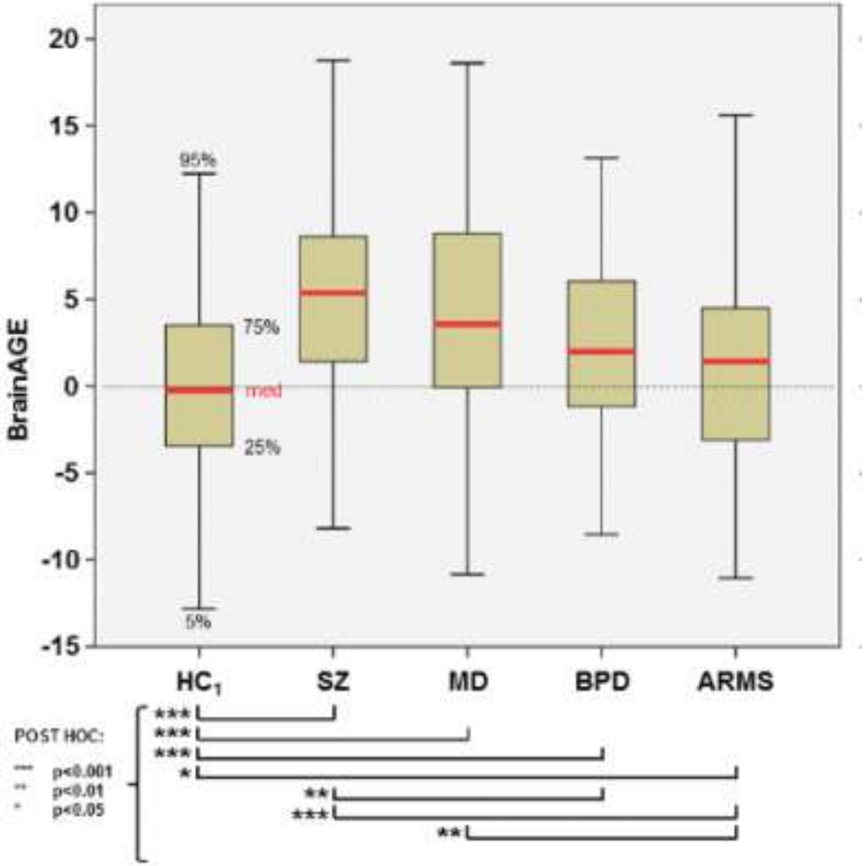
- **Franke et al. 2012** propose that the difference between estimated and chronological age might be clinically informative
  - **BrainAGE** = Brain Age Gap Estimation
  - Applied to 394 subjects, 5-18 years
  - Features derived from gray / white matter maps (after tissue segmentation)
  - Classification of children (5-10 years) vs. adolescents (13-18 years) 97% accurate
  - Correlation with predicted age r=0.93; average error 1.1 years
  - Pre-terms (GA<27 weeks) had significantly decreased BrainAGE (-1.96 years)



Child vs. adolescent based on BrainAGE vs. volumes

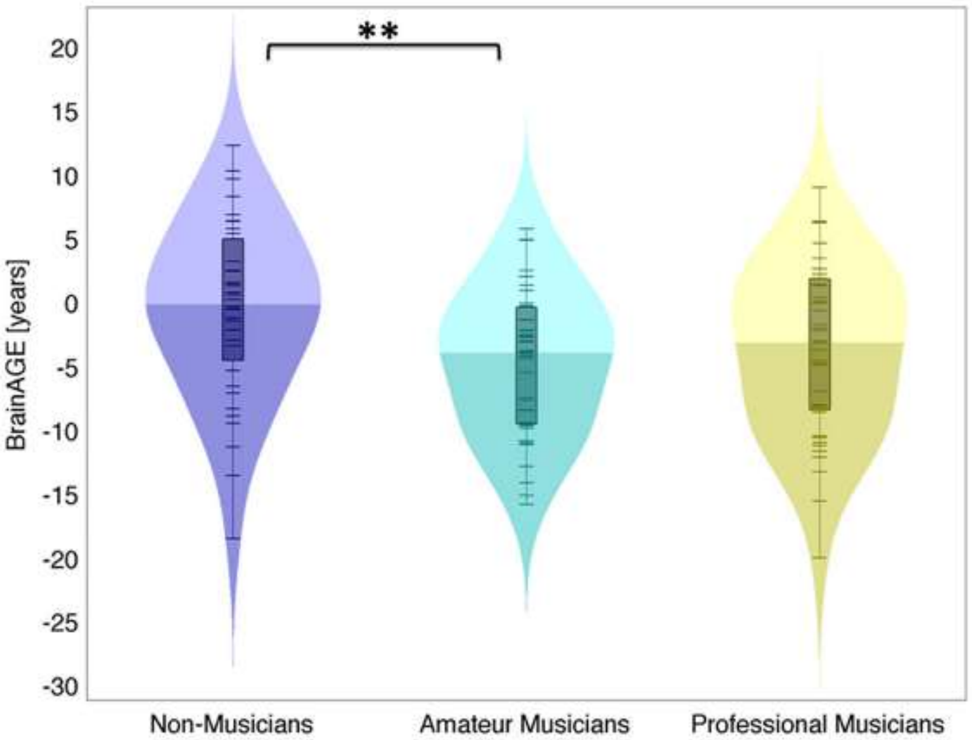
BrainAGE in Psychiatric Disorders

- **Koutsouleris et al. 2013** use SVR on GM maps to train a BrainAGE model on 800 healthy controls, age 18-65
  - Mean absolute BrainAGE 4.6 yrs
  - Applying the model to patients with psychiatric disorders indicates accelerated aging
    - Schizophrenia +5.5 yrs
    - Major depression +4.0 yrs
    - Borderline PD +3.1 yrs
    - At-risk mental state +1.7 yrs



Keeping Your Brain Young with Music

- **Rogenmoser et al. 2018** compare BrainAGE in 42 professional and 45 amateur musicians to 38 non-musicians
  – Matched w.r.t. age ≈25±4 years, gender, engagement in sports
  – BrainAGE model trained on an independent sample of 631 subjects
  – Making music was found to have a significant effect on BrainAGE
  – Average BrainAGE in musicians was 4.12 years lower than in non-musicians



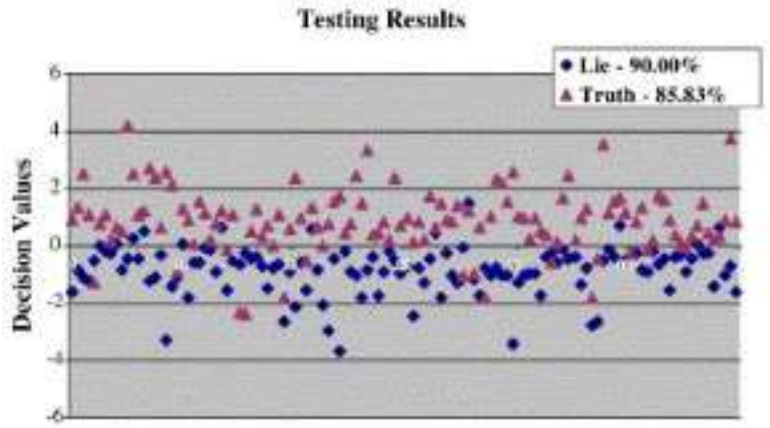
Summary: Applications in Brain Diseases

- The use of **learned classifiers** to support the diagnosis of brain diseases based on neuroimaging seems feasible in principle, but is currently limited by available dataset sizes and heterogeneity
  – Technical challenges include model uncertainty and interpretability, computational effort
- **BrainAGE** models of brain maturity can be trained on a representative set of healthy subjects and have shown systematic effects of premature birth or psychiatric disorders

9.3 Applications in “Mind Reading”

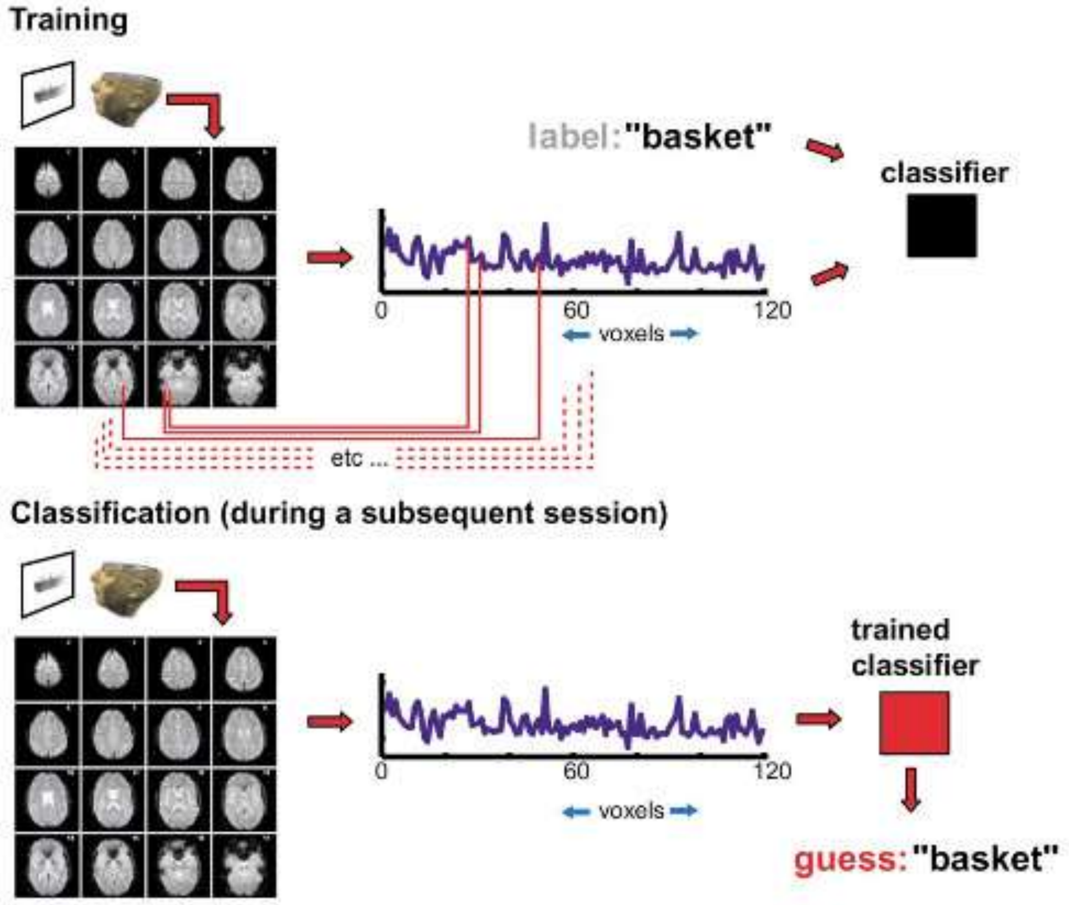
Detecting Lies

- **Davatzikos et al.** 2005 report 88% accuracy in detecting “lies” in an fMRI scanner
  - Each of 22 subjects is given two playing cards
  - Many cards are shown in the scanner, subjects press one button if they have it, another one if they don’t
  - Subjects are asked to „lie“ about possession of one of their cards (they decide which one)
  - GLM with canonical hemodynamic response to estimate activation map after each card
  - Features computed by subdividing the brain into 560 cubes and averaging activations over them
  - SVM with leave-one-subject-out cross-validation



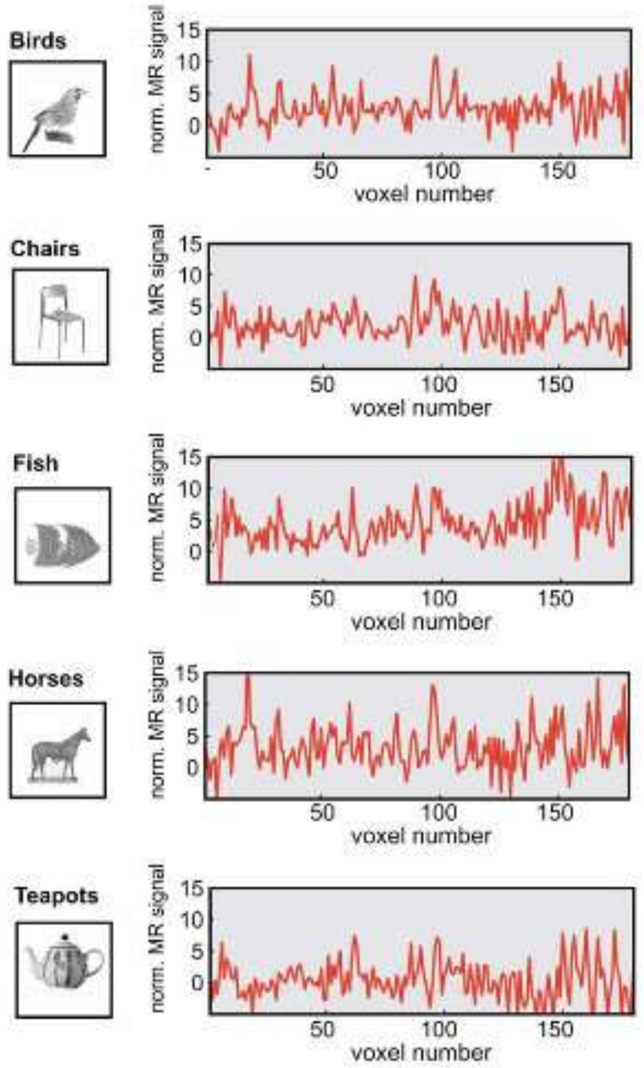
Detecting What Subjects See

- **Cox et al. 2003** estimate from fMRI data from which category (out of ten) a subject is watching an image (during 20 sec)
  - **Training** based on data from the same subject, but taken in a different session
  - At **test time,** showing either the same or other images from the same category



Detecting What Subjects See: Features

- **Preprocessing:** Motion correction, no smoothing, no spatial normalization
  - **Thermoplastic head-restraint** for reproducible head placement and to minimize motion
- **Features computed** by averaging voxel intensities over 20 sec and removing the mean from a control state
- **Features selected** by F-test (difference between categories); additional features from presenting whole vs. scrambled objects



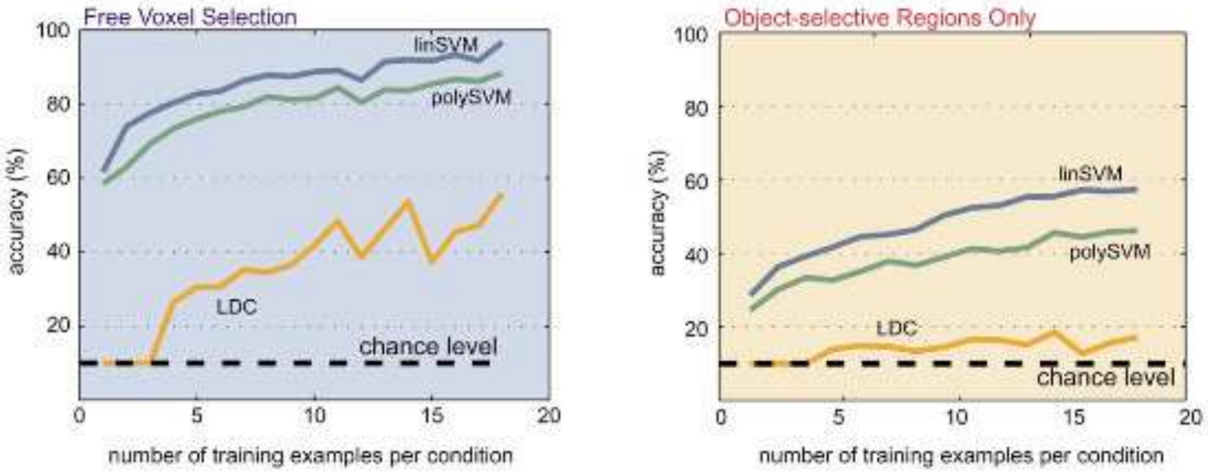
Detecting What Subjects See: Results

- Prediction much better than random chance (i.e., 10%)

- Relatively little training data suffices

	free voxel subset selection			restricted voxel selection		
10-way, same exemplars	LDC	linear SVM	poly SVM	LDC	linear SVM	poly SVM
S1 (5 sessions, 56 blocks/cond)	51 %***	85 %***	83 %***	15 %*	41 %***	41 %***
S2 (5 sessions, 40 blocks/cond)	29 %***	58 %***	52 %***	14 %*	33 %***	28 %**
10-way, different exemplars						
S1 (4 sessions, 40 blocks/cond)	29 %**	59 %***	53 %***	13 %	29 %**	19 %*
S2 (2 sessions, 20 blocks/cond)	42 %***	82 %***	80 %***	14 %	37 %**	29 %**
S3 (2 sessions, 20 blocks/cond)	52 %***	90 %***	93 %***	21 %*	55 %***	49 %***
S4 (2 sessions, 20 blocks/cond)	56 %***	97%***	88 %***	21 %*	55 %***	49 %***

Classifier Accuracy as a Function of Number of Training Examples per Condtion Used

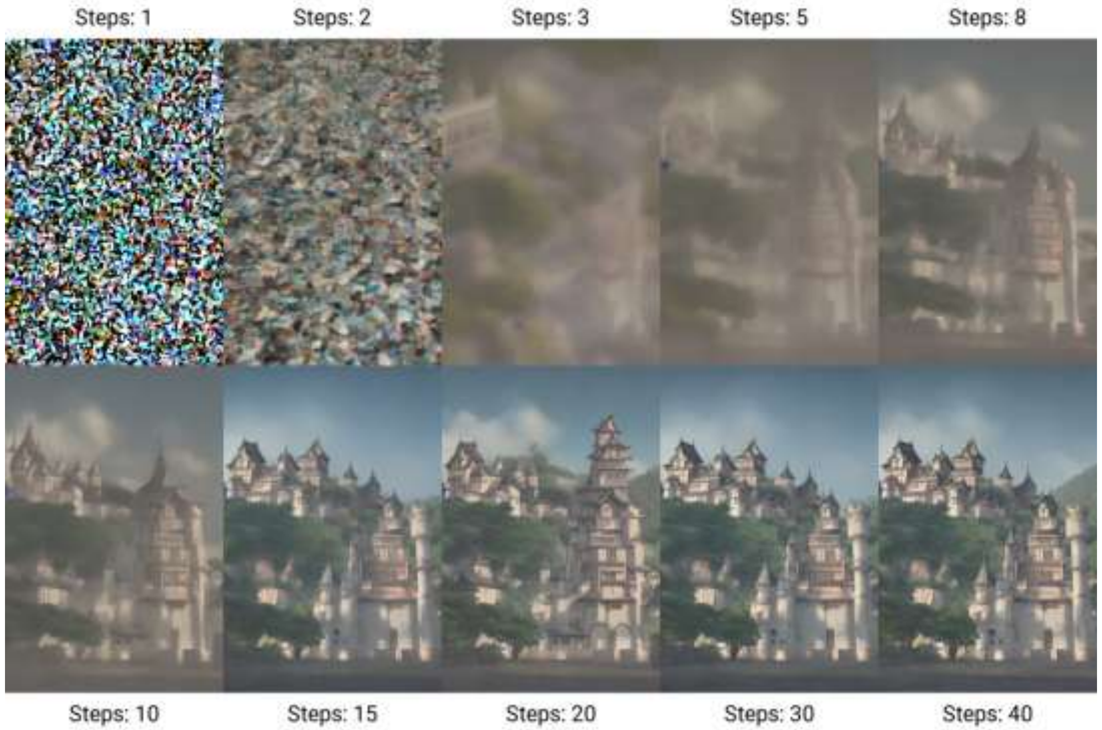


Reconstructing What Subjects See

- We will look at a recent work on reconstructing images based on fMRI scans [Takagi/Nishimoto, CVPR 2023] that makes use of generative AI
- [Naselaris et al., Neuron 2009] previously presented an approach that was based on very similar ideas
  – Extensive prior knowledge about structure of natural images
  – Constrain image generation with semantic information from higher visual areas

Diffusion Models

- **Diffusion models** are a generative technique. They are based on gradually adding noise to training examples and learning a model that removes it again. This way, they learn to generate meaningful signal (e.g., images) from noise.

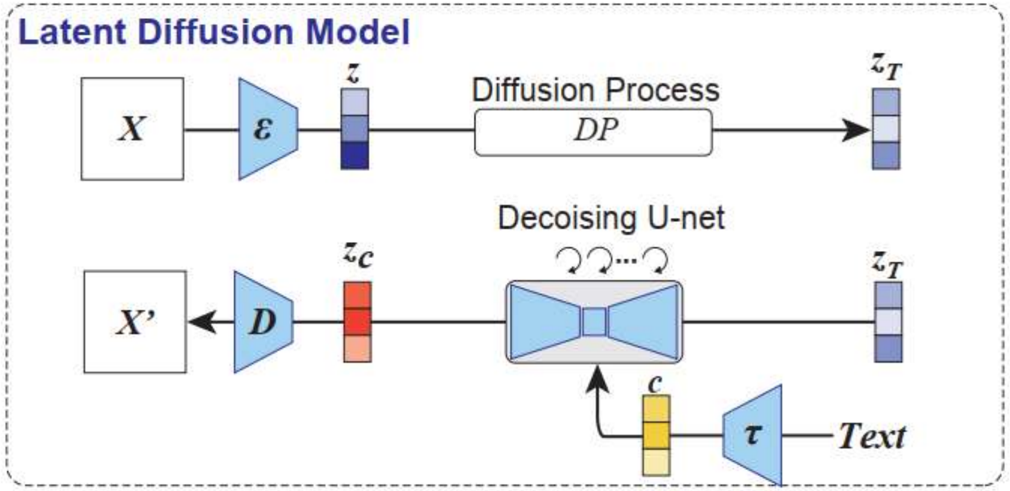


Images: benlisquare via Wikipedia

Use of Text Embeddings in Diffusion Models

- **Text embeddings** represent text as vectors
  - Similar vectors for semantically similar text
- Giving the denoising process access to an embedding of a textual description of the image contents helps it to recover the original image (during training) and allows us to specify desired image contents afterwards

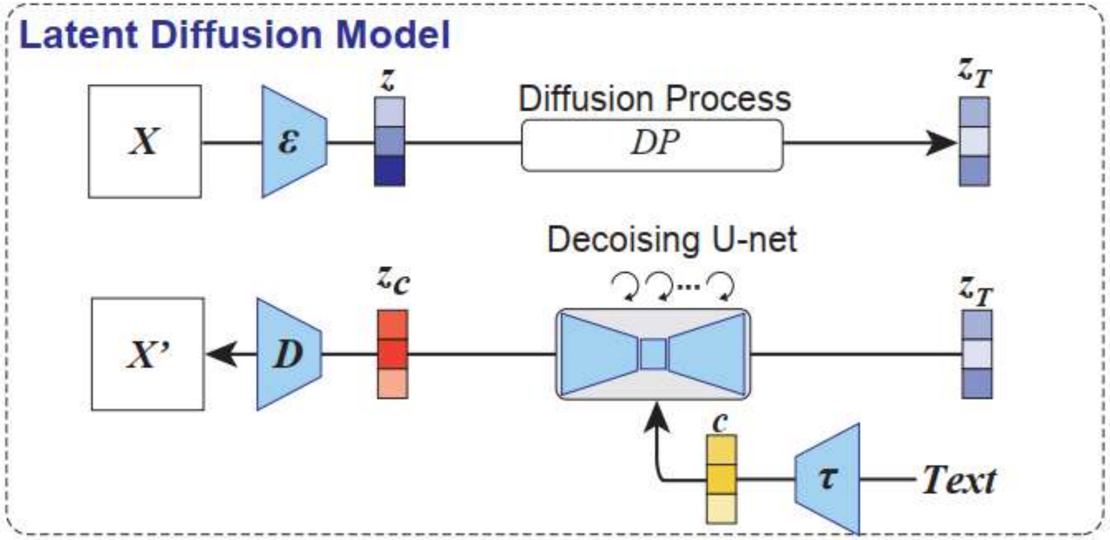
Prompt: "a photograph of an astronaut riding a horse"



Images: Wikipedia / [Takagi/Nishimoto, 2023]

Latent Diffusion Models

- **Variational autoencoders** are another generative technique. They encode (compress) signals into a lower-dimensional latent space, and decode (reconstruct) them again
  - Regularization ensures that similar latent codes yield similar images
- **Latent diffusion models** create high-resolution output at a reduced computational cost by operating within the latent space of an autoencoder
  - *Intuition*: Diffusion should care about semantics, not small, imperceptible details

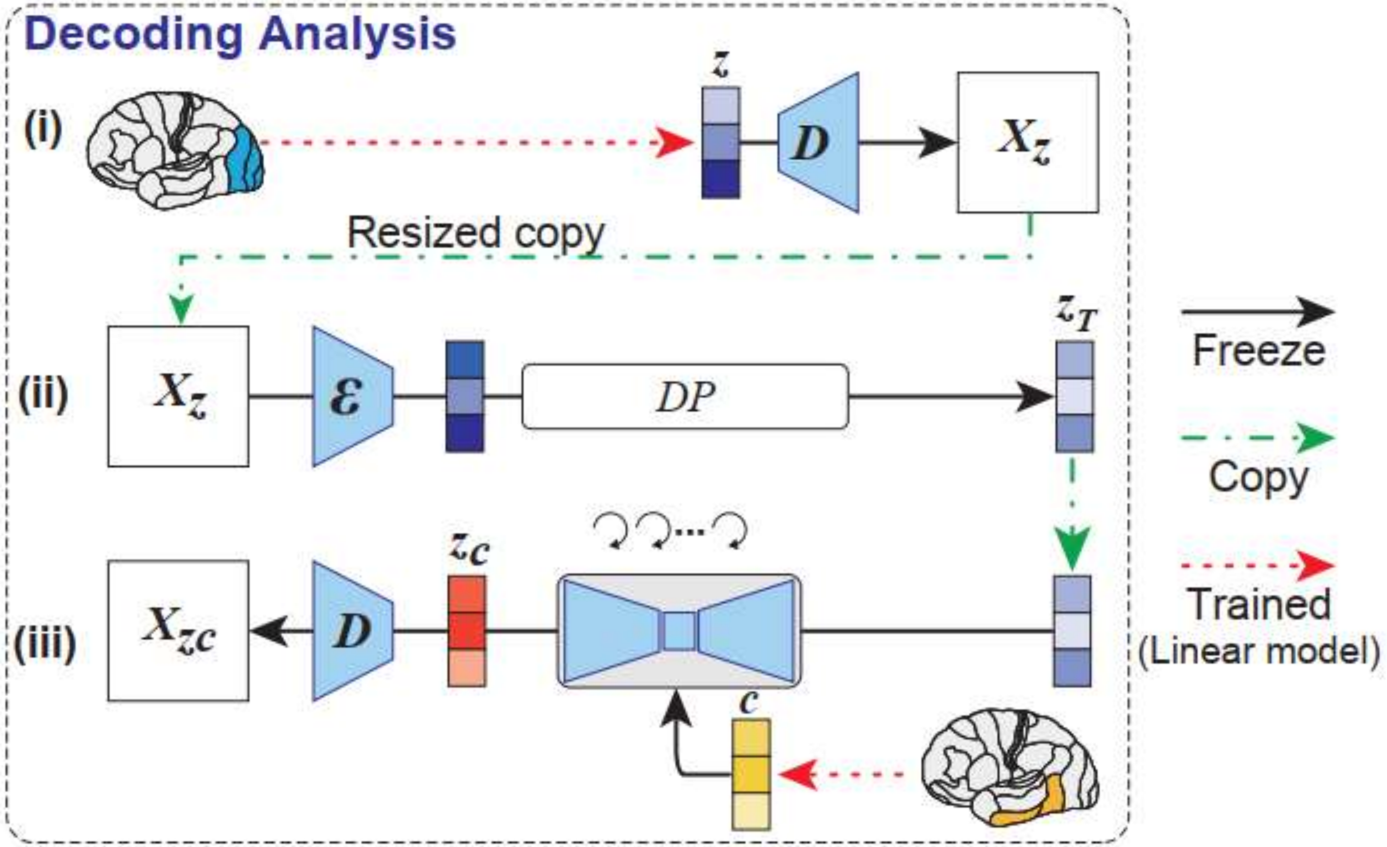


Images: [Takagi/Nishimoto, 2023]

Decoding Activations via Diffusion Models

Suggested approach in [Takagi/Nishimoto, 2023]:

1. Linearly transform fMRI activations in early visual cortex to an initial image embedding z, decode to a coarse picture
2. Encode upsampled image, add noise to obtain z_T
3. Linearly transform fMRI activations in higher-level visual cortex to a text embedding c
4. De-noise z_T conditioned on text embedding c to final image embedding z_c

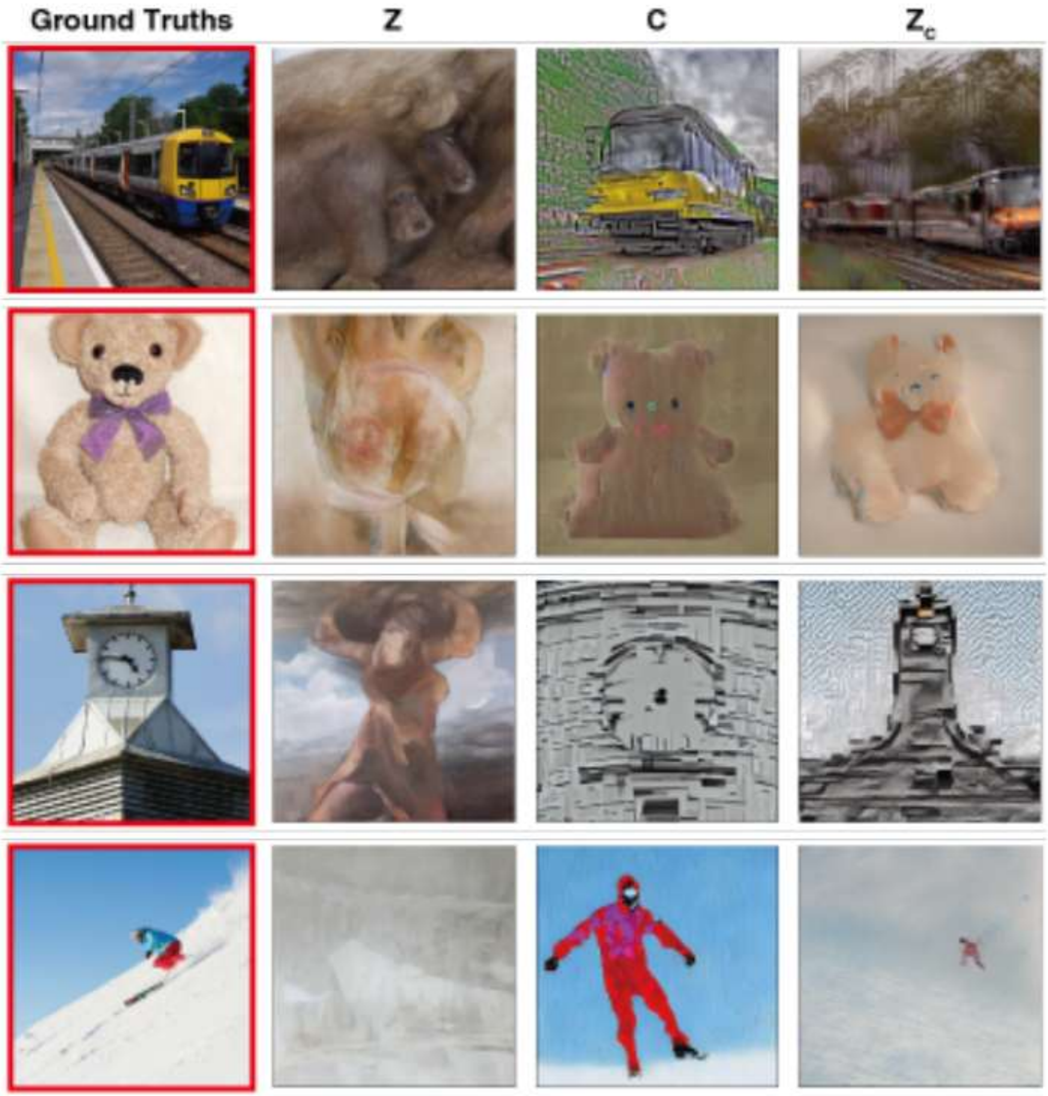


Images: [Takagi/Nishimoto, 2023]

Results

Columns show:

1. Image seen by test subject
2. Initial, coarse reconstruction
3. Reconstruction using only the text embedding
4. Reconstruction with the full proposed pipeline



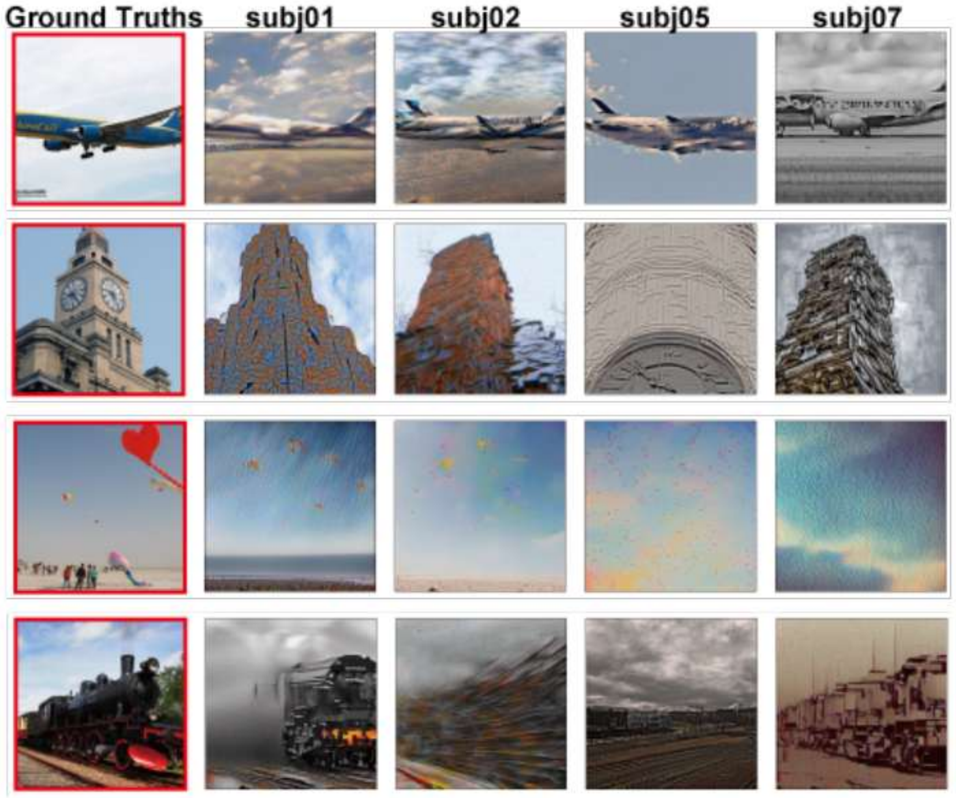
Images: [Takagi/Nishimoto, 2023]

Limitations

Results are impressive given that they are based on simple linear transformations from fMRI activation maps to the latent space of a generative model, which itself has not been optimized for this task.

However:

- Mapping needs to be trained per-subject and reconstruction quality differs between subjects. Focus is on one subject where it worked very well
- Relies on average of three fMRI runs
- In all cases, five candidate reconstructions were created, only the one closest to the real image is shown



Images: [Takagi/Nishimoto, 2023]

Summary: Machine Learning in Neuroimaging

Machine learning was successfully used for a range of classification / regression tasks in neuroimaging:

- Diagnosing disease (not yet at a clinically useful level)
- Estimating BrainAGE
- Detecting deception (in very simple, controlled tasks)
- Detecting what class of images subjects are viewing

To a certain extent (and with per-subject training), **generative models** can be used to “decode” fMRI activation maps, e.g., into images

Exam-Like Questions

1. State two advantages of Support Vector Machines that explain their popularity for supervised learning from neuroimaging data.
2. Propose a feature representation of a structural MRI scan that is suitable for BrainAGE estimation.

Further Reading

- Bernhard Schölkopf, Alexander J. Smola: *Learning with Kernels.* MIT Press, 2002
- Russell A. Poldrack, Jeanette A. Mumford, Thomas E. Nichols: *Handbook of Functional MRI Data Analysis.* Cambridge University Press, 2011
- Lauren O'Donnell, Thomas Schultz: *Statistical and Machine Learning Methods for Neuroimaging: Examples, Challenges, and Extensions to Diffusion Imaging Data.* In: *Visualization and Processing of Higher Order Descriptors for Multi-Valued Data,* pages 299-319, Springer, 2015

Announcements

- I will offer a **Q&A session** on Thu Jan 29
  - Please prepare questions!
- The DZNE scanner tour on Monday, February 2, will start at **13:00 sharp**
  - Please arrive a few minutes earlier and enter the building as a group
  - Annika Mikliss will be there as contact person
  - Space is tight, join only if you managed to book a spot on eCampus
  - Venusberg-Campus 1/99, Bus stop Uniklinikum Süd
- Please participate in the **lecture evaluation** via https://fha8.de/ZMMGX

Reminder: Written Exams

- Our **written exam** takes place
  - On February 5, 2026 between 14:00-16:00
  - In Lecture Hall IX, Am Hof 1 / Regina-Pacis-Weg 1
- The **second attempt** takes place
  - On March 10, 2026 between 10:00-12:00
  - In Lecture Hall II, Meckenheimer Allee 176